

Sri Lakshmi Vempati

Vempatisrilakshmi33@gmail.com | 530-863-0030 | <https://www.linkedin.com/in/srilakshmivempati>

PROFESSIONAL SUMMARY

Computer Engineering graduate student specializing in digital circuits, system design, and verification, DFT and actively looking for summer/Fall 2024,2025 internship opportunities.

EDUCATION

California State University, Sacramento, Sacramento, CA

Master of Science in Computer Engineering

GPA: 3.8

Bharath Institute of Higher Education and Research, Chennai, India

Bachelors in Computer Science and Engineering

GPA: 3.7

COURSE WORK

Adv logic design	Digital ICs
Micro-computer system design	Computer system structures
System Design and Verification	Adv micro-architecture
DFT	Adv timing Analysis

TECHNICAL SKILLS

Programming Languages: Assembly language, C, Python, Verilog, System Verilog, Perl.

Technologies: Analog & Digital Design, VLSI, PCIe, DRAM, Arm processor.

Tools: Quartus Prime, Cadence Virtuoso, FPGA(DE-10Lite), Model Sim, Synopsis VCS, Tessent.

OS: Linux/Unix.

ACADEMIC PROJECTS

Design, simulation, and layout of 4-bit ALU

Tool used: Cadence Virtuoso

- Designed and implemented 4-bit CMOS ALU using 45nm technology in Cadence.
- Designed and implemented various circuit schematics and layouts with LVS and DRC with zero errors and also performed post-extraction.
- Performed operations like Bitwise Addition, Subtraction, XOR, NOT, and Multiplication.

Design of 4 port Switch Router

- Designed and verified a 4-port switch router using System Verilog (SV) to efficiently route data packets between network ports based on destination addresses.
- Developed a comprehensive UVM-based verification environment, including testbenches for functional verification and coverage analysis to ensure design accuracy and reliability.
- Implemented RTL design and simulation techniques, utilizing assertion-based verification to validate performance and optimize resource utilization.

Simulation of Digital Circuits

Tool used: Quartus prime

- Simulated digital circuits like shift registers, sequential multipliers, Mealy and Moore sequence detectors, Traffic light signal controllers, clock dividers, and Vending machines.
- The results are displayed and validated using the DE-10 Lite board.

Designed and verified FIFO

Tool used: synopsis VCS

- Designed a parametrizable FIFO buffer using System Verilog, the FIFO was configurable to support different data widths.
- Verified implementation using a reusable OOPS Env and created a suite of tests that test various functionalities of the DUT.

Design and Verification of DMA Controller Using the Synopsis

Tool used: synopsis VCS

- Designed the core DMA Controller module for managing different data transfers such as addressing and data moment control.
- Developed using specifications like memory, peripheral interfaces, Transfer modes, arbitration, prioritization, and also burst management.
- Design & verify all these scenarios with different test cases.

Cache controller simulator | c language

- Design and implement an N-way where (N=1,2,4,8,16) set associative cache memory with a cache controller using C language to optimize memory access and improve system performance.
- Developed the cache Size controller(16k,32k,64k,128k) and block size is 32,64,128 and 256 in C to manage cache operations such as read, write, hit, and miss.
- Implemented algorithms for cache replacement policies (e.g., LRU, FIFO).

PUBLICATIONS & ACHIEVEMENTS

- Received a badge and certification from Siemens for the on-demand course on DFT.
- Course completion certificate from the I Master Technologies in Python and Machine Learning.
- Published paper on semi-supervised ML model for probe attack on IJSDR 2023.

